Theorem

Let $T: V \longrightarrow V$ be a linear operator, and let S be a (finite) basis of V. Then, for any vector $v \in V$,

$$[T]_S [v]_S = [T(v)]_S$$

Proof: Given $T: V \longrightarrow V$ is a linear operator.

S be a basis of V

Suppose $S = \{u_1, u_2, \ldots, u_n\}$

$$T(u_i) = a_{1i}u_1 + a_{2i}u_2 + \cdots + a_{ni}u_n$$

$$= \sum_{j=1}^{n} a_{ji} u_j$$

Then $[T]_S$ is the nth square matrix whose ith row is, $(a_{j1} \quad a_{j2} \quad \cdots \quad a_{jn})$

Now, suppose,

$$v = k_1u_1 + k_2u_2 + \cdots + k_nu_n = \sum_{i=1}^{n} k_i u_i$$

$$\therefore [v]_S = (k_1 \quad k_2 \quad \cdots \quad k_n)^T$$

Now jth entry of $[T]_S [v]_S$ is obtained by multiplying jth row of $[T]_S$ and $[v]_S$

i.e. $a_{j1}k_1 + a_{j2}k_2 + \cdots + a_{jn}k_n$

Now, $T(v) = T \sum_{i=1}^{n} (k_i u_i)$

$$= \sum_{i=1}^{n} k_i T(u_i)$$

$$= \sum_{i=1}^{n} k_i \left(\sum_{j=1}^{n} a_{ji} u_j \right)$$

$$= \sum_{j=1}^{n} \left(\sum_{i=1}^{n} a_{ji} k_i \right) u_j$$

$$= \sum_{j=1}^{n} (a_{j1}k_1 + a_{j2}k_2 + \cdots + a_{jn}k_n) u_j$$

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